DECK 10 · CORPORATE ISSUERS

Working Capital, Capital Allocation & Business Models

Short-term cycle, long-term investment, and the business model underneath both

CFA Level I · Vol 3 Learning Modules 4, 5, 7

Roadmap — Working Capital, Capital Allocation & Business Models

3 sections · ~2 hours · Built bottom-up from CFA readings

01

Working capital & liquidity

CCC mechanics · Inditex & discount retailers · supplier financing · liquidity ratios · drags & pulls · conservative vs aggressive

02

Capital investments & allocation

project types · 4-step process · NPV / IRR / ROIC · inflation · cognitive & behavioral pitfalls · real options with decision tree

03

Business models

five Ws framework · eight archetypes · pricing taxonomy · platforms & network effects · B2B vs B2C · subscription unit economics (CAC/LTV)

Learning objectives

By the end of this deck you should be able to:

1. Compute a cash conversion cycle from activity ratios, interpret negative CCCs, and price supplier trade credit against a bank loan.

2. Evaluate firm liquidity using current / quick / cash ratios and distinguish primary from secondary liquidity sources, including drags and pulls.

3. Apply NPV, IRR and ROIC to capital-investment decisions and recognise the cognitive and behavioral biases that distort real-world capital allocation.

4. Identify the four types of real option (timing, sizing, flexibility, fundamental) and compute a project NPV that embeds a decision tree.

5. Decompose a business model into its five core questions, classify it into one of eight conventional archetypes, and compute the unit economics of a subscription business (CAC, LTV, payback).

SECTION 01

Working capital & liquidity

The operating cycle converts raw materials into cash, and the cash conversion cycle is the gap left over after supplier financing

DOH lengthens the cycle, DSO lengthens the cycle, DPO shortens it. Negative CCC means the firm collects before it pays.

$$CCC = DOH + DSO - DPO$$

where: DOH = days of inventory on hand · DSO = days sales outstanding · DPO = days payable outstanding. Units: days.

Industry (2020 S&P 1500 avg)	CCC (days)	Driver
Pharmaceuticals	180	Large regulatory inventory stockpile
Apparel	123	Seasonal inventory, long DOH
Semiconductors	121	Complex multi-stage inventory
All S&P 1500 companies	78	Benchmark
Auto & auto parts	66	Shorter DOH via JIT manufacturing
Retail	61	Card sales, short DSO
Oil & Gas	40	—
Software	22	Prepaid subscriptions
Media	12	Subscription & ad revenue upfront

Inditex runs a structurally negative cash conversion cycle — customers pay before suppliers are paid

Fast-fashion turnover + point-of-sale cash/card payments + huge supplier base → ZARA's vertically integrated model funds growth from suppliers.

Metric (EUR millions unless days)	2021	2020
Accounts receivable	842	715
Inventories	3,042	2,321
Accounts payable	6,199	4,659
Sales	27,716	20,402
Days sales outstanding (DSO)	11 d	13 d
Days of inventory on hand (DOH)	93 d	94 d
Days payable outstanding (DPO)	190 d	189 d

CCC & NET WORKING CAPITAL COMPUTATION

Net working capital 2021 = 842 + 3,042 − 6,199 = −EUR 2,315M. As % of sales: −2,315 / 27,716 = −8.4%.

CCC = 11 + 93 − 190 = −86 days · Inditex is financed by its supply chain, not the reverse.

Source: CFA Institute, Curriculum Vol 3, Learning Module 4, Inditex case study, pp. 103–108.

Five large US discount retailers split neatly into negative-, zero- and positive-CCC camps depending on payment terms and perishability

All five accept card payment at the register, so DSO is tiny. The spread is driven by DOH (perishables vs apparel) and supplier leverage.

Retailer (2021)	DSO	DOH	DPO	CCC = DOH + DSO − DPO
Walmart	5 d	48 d	47 d	5 + 48 − 47 = +6 d
Target	2 d	68 d	75 d	2 + 68 − 75 = −5 d (shortest)
Costco	3 d	29 d	34 d	3 + 29 − 34 = −2 d
TJX	7 d	63 d	47 d	7 + 63 − 47 = +23 d (longest)
Ross Stores	2 d	61 d	63 d	2 + 61 − 63 = 0 d

WHY THE SPREAD

Costco sells ~54% food/perishable goods — inventory must be sold before it spoils, driving DOH down to 29 days. TJX (off-price apparel) carries the most inventory across narrower supplier relationships. Target leans hardest on supplier financing — 75 days payable is the highest DPO of the five.

Source: CFA Institute, Curriculum Vol 3, Learning Module 4, Example 2, pp. 105–106.

Forgoing a 2/10 net 30 supplier discount is an implicit 44.6% annual loan — almost always cheaper to borrow from a bank

The borrower is effectively paying the discount to use the supplier's money for (payment period – discount period) days. Annualize to compare.

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EAR = [ 1 + Disc% / (100% – Disc%) ]^(365 / (net days – disc days)) – 1
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where: Disc% = prompt-payment discount · net days = total payment period · disc days = discount period.

KEOWN CORP — 2/10 NET 30 vs 7.7% BANK LOAN

Supplier offers 2% discount if paid in 10 days, otherwise net in 30. Bank loan EAR = 7.7%.

1. Base ratio: 2% / (100% – 2%) = 0.02041
2. Periods / year: 365 / (30 – 10) = 18.25
3. EAR = (1.02041)^18.25 – 1 = 44.6%
4. 44.6% supplier cost ≫ 7.7% bank loan → borrow from bank, take the discount.

Supplier financing EAR = 44.6% → USE THE BANK

Trade credit terms	Implied EAR
1/10, net 50	9.6%
1/14, net 30	25.8%
3/15, net 60	28.0%
2/15, net 40	34.3%
2/10, net 30	44.6%

Source: CFA Institute, Curriculum Vol 3, Learning Module 4, Example 1 Keown Corp, pp. 104–105.

Current, quick, and cash ratios rank firms by progressively more conservative definitions of "what can pay my bills right now"

The three ratios share a denominator. The numerator shrinks as the definition of "liquid" tightens. Order is always: Current ≥ Quick ≥ Cash.

CURRENT RATIO

Current assets / Current liabilities.

Broadest. Includes inventory and receivables. Good for going-concern liquidity.

QUICK RATIO

(Cash + ST marketable + AR) / Current liabilities.

Excludes inventory (slow to liquidate). "Acid test" for stress scenarios.

CASH RATIO

(Cash + ST marketable) / Current liabilities.

Most conservative. Assumes AR cannot be collected in crisis. Crisis-liquidity metric.

Licht Vernieuwend N.V. (EUR M)	20X1	20X2	Δ	
Cash ratio = (Cash+Sec) / CL	0.60	0.50	−0.10	worse
Quick ratio = (Cash+Sec+AR) / CL	1.00	0.89	−0.11	worse
Current ratio = CA / CL	1.36	1.52	+0.16	better
Verdict based on most conservative measure				**Cash ratio down → less liquid**

Source: CFA Institute, Curriculum Vol 3, Learning Module 4, §3 "Liquidity", pp. 117–122.

Drags on liquidity slow cash in; pulls on liquidity accelerate cash out — both shorten the runway

Primary liquidity sources (cash, CFO, revolver) are cheap. Secondary sources (asset sales, equity raise, bankruptcy) signal distress.

DRAGS ON LIQUIDITY (inflows slow)

- Uncollected receivables (delinquencies)
- Obsolete inventory
- Tighter borrowing constraints
- Customer credit-quality deterioration

PULLS ON LIQUIDITY (outflows accelerate)

- Early payments with no discount
- Reduced supplier credit limits
- Shrinking short-term credit lines
- Chronically low liquidity positions

Liquidity source	Examples	Signal
Primary — cash & securities	Bank balances, T-bills, commercial paper	Healthy
Primary — borrowings	Bank revolver, trade credit, CP issuance	Healthy
Primary — cash flow from ops	The long-run engine; tracked via statement of cash flows	Healthy
Secondary — cut CAPEX / dividend	Delta deferred USD 500M CAPEX in 2020	Stress
Secondary — issue equity	Singapore Airlines SGD 8.8B rights issue, 2020	Stress
Secondary — sell assets	Convertible bond, asset sale, spin-off	Distress
Secondary — bankruptcy	Chapter 11 / insolvency reorganisation	Distress

Working-capital policy sits on a spectrum from conservative to aggressive — each lever has a cost and a risk

Conservative = safer but costlier. Aggressive = cheaper but exposed to rollover risk. Moderate = match-funding — the IB default.

Dimension	Conservative	Moderate ("matched")	Aggressive
Current-asset level vs sales	Larger	Moderate	Smaller
Funding mix	More long-term debt + equity	Match permanent→LT; variable→ST	More short-term debt
Financing cost	Higher	Medium	Lower
Rollover / refinancing risk	Minimal	Limited to variable share	High — hit hardest in crises
Flexibility in market stress	High	Medium	Low
Typical user	Early-growth, regulated	Mature diversified	Mature low-margin, stable demand

WHEN AGGRESSIVE BITES

The 2008–09 crisis and early 2020 COVID shock repeatedly broke aggressive WC firms: inability to roll CP or bank lines forced equity raises at distressed prices. Match-funding is the defensive default for any firm with cyclical revenues.

Source: CFA Institute, Curriculum Vol 3, Learning Module 4, §4 Exhibits 10–13, pp. 123–127.

SECTION 02

Capital investments & allocation

Capital investments fall into four categories that differ in risk, return, and discretion — not all are elective

Regulatory projects can proceed with negative NPV (the alternative is a bigger fine). The other three are all discretionary.

Category	Purpose	Risk	Example
Going concern (maintenance)	Replace aging assets; keep lights on	Low	Data-center cooling upgrade
Regulatory / compliance	Comply with rules; avoid fines	Low	Danske AML remediation programme
Expansion of existing business	Scale up or extend scope; adjacent products	Med.	Sony Interactive acquires Bungie (USD 3.6B, 2022)
			Kirin (brewer) buys

MAINTENANCE PROXY

Analysts commonly estimate annual maintenance CAPEX = D&A expense, best accurate for short-lived assets.

Expansion CAPEX ≈ Total CAPEX − D&A.

FINANCING RULE OF THUMB

Match the debt tenor with the asset life. Utility replacing 30-year turbines issues 30-year bonds.

Rollover risk = shorter debt than asset life. Negative carry = longer debt than needed.

Source: CFA Institute, Curriculum Vol 3, Learning Module 5, §2 "Capital Investments", pp. 139–145.

The capital allocation process is a four-step loop from idea generation to post-investment review — most pitfalls live between steps 2 and 3

NPV and IRR are tools used only in step 2. Step 4 — post-audit — is the step firms most often skip and the step analysts most often value.

1

1. Idea generation

Managers, business-unit heads, consultants, M&A bankers propose projects. Ideas usually come from within existing operations (inertia risk — see later).

2

2. Investment analysis

Model after-tax cash flows with realistic assumptions. Compute NPV and IRR. Apply the required rate of return appropriate to project risk — NOT to financing source.

3

3. Planning & prioritisation

Rank projects; impose portfolio constraints (financing, strategic fit, ESG). Return surplus capital to shareholders when value-creative projects are exhausted.

4

4. Monitoring & post-audit

Track actuals vs projections. Identify systematic optimism. Feed lessons back into step 1. Exercise real options (expand, abandon) as information arrives.

Source: CFA Institute, Curriculum Vol 3, Learning Module 5, §3 Exhibit 3, pp. 146–147.

NPV discounts every after-tax cash flow at the hurdle rate; accept iff NPV ≥ 0, and it beats IRR whenever they disagree

NPV is in currency units — interpretable as the dollar increase in shareholder wealth. IRR is in rate units — useful for ranking but not for choosing.

$$NPV = CF_0 + \Sigma \; CF_t / (1 + r)^t \qquad \text{Accept iff } NPV \geq 0$$

where: CF_0 = initial outlay (typically negative); r = required return on similarly risky investment; CF_t = after-tax cash flow at t.

GERHARDT CORP — EUR 50M FACILITY, r = 10%

$CF_0 = -50$; $CF_1 \ldots CF_4 = +16$ each; $CF_5 = +20$. All amounts in EUR millions.

1. $PV(y1) = 16 / 1.10 = 14.545$ $\quad PV(y2) = 16 / 1.10^2 = 13.223$
2. $PV(y3) = 16 / 1.10^3 = 12.021$ $\quad PV(y4) = 16 / 1.10^4 = 10.928$
3. $PV(y5) = 20 / 1.10^5 = 12.418$
4. $NPV = -50 + 14.545 + 13.223 + 12.021 + 10.928 + 12.418 = +13.136$

NPV = +EUR 13.136M → ACCEPT. Project adds ≈ 26% to the outlay in present value.

Source: CFA Institute, Curriculum Vol 3, Learning Module 5, Example 4 Gerhardt Corp NPV, pp. 147–149.

Inflation-adjustment is a matching problem: discount nominal cash flows at a nominal rate, real cash flows at a real rate — never mix

Both approaches produce the same NPV when done correctly. The error is silently mixing conventions, which inflates (or deflates) NPV spuriously.

$$(1 + r_{nominal}) = (1 + r_{real}) \times (1 + \pi) \quad \Rightarrow \quad r_{real} = (1 + r_{nominal}) / (1 + \pi) - 1$$

where: π = expected inflation. Mix conventions (nominal CF at real r) and the result is meaningless.

Project year	t=0	t=1	t=2	t=3	NPV
Real CF (today's prices, π = 5%)	−100.0	40.0	40.0	40.0	—
Nominal CF = Real × $(1.05)^t$	−100.0	42.0	44.1	46.3	—
Discount at nominal r = 10%	—	38.18	36.45	34.79	+9.43
Discount real CF at real r = 4.76%	—	38.18	36.45	34.79	+9.43
Both approaches → identical NPV when matched					**+EUR 9.43**

Source: CFA Institute, Curriculum Vol 3, Learning Module 5, Inflation principle + ScolarCorp Q&A, pp. 160–161.

IRR is the rate that zeros NPV — intuitive, but it reinvests cash flows at itself and breaks down when signs alternate

Gerhardt IRR = 19.52% on the same cash flows as the previous slide. Accept iff IRR ≥ hurdle. But the rule is fragile.

$$\text{IRR} = \text{rate } r^* \text{ such that } \Sigma_t \; CF_t / (1 + r^*)^t = 0$$

FAILURE 1 · MULTIPLE IRRs

Signs flip more than once (−, +, −). CFs −1,000 / +5,000 / −6,000 solve NPV = 0 at BOTH r = 100% and r = 200%.

Excel reports the lowest root only.

FAILURE 2 · SCALE MISMATCH

Project A: invest 100, earn 150 (IRR 50%). Project B: invest 1,000, earn 1,300 (IRR 30%).

A has higher IRR but B adds more NPV. If mutually exclusive, pick B.

FAILURE 3 · REINVESTMENT

IRR assumes interim cash flows earn IRR forever. A 30% IRR project assumes reinvestment at 30% — rarely true.

NPV uses the cost of capital instead. Theoretically superior.

[1] IRR reinvestment assumption: interim CFs earn the IRR. NPV implicitly reinvests at r — usually more realistic above WACC.

[2] For mutually exclusive projects of different scale, the higher NPV always wins — not the higher IRR.

Source: CFA Institute, Curriculum Vol 3, Learning Module 5, pp. 150–153 (IRR and limitations).

ROIC is the only external-analyst tool for gauging whether a firm earns its cost of capital — decompose it into margin × turnover

Unlike NPV and IRR (which need project-level data), ROIC can be computed from public financial statements. Compare ROIC vs WACC = value test.

ROIC = (1 – t) × Operating profit / Avg. invested capital = After-tax margin × Capital turnover

where: Invested capital = LT liabilities + equity (average of beginning and ending). Working capital excluded.

ROIC COMPUTATION — HYPOTHETICAL ISSUER, YEAR 2

After-tax operating profit = 24,395. LT debt + Share capital + RE: Y1 = 276,387; Y2 = 282,280.

1. Avg invested capital = (276,387 + 282,280) / 2 = 279,333
2. ROIC = 24,395 / 279,333 = 8.73%
3. Required return (hurdle) = 10%
4. 8.73% < 10% → firm destroys value — unless reversed

ROIC 8.73% < hurdle 10% → VALUE DESTRUCTION

DUPONT DECOMPOSITION

ROIC = (After-tax profit / Sales) × (Sales / Invested capital)
= Margin × Turnover

Two paths to a high ROIC:

· Premium strategy — high margin, low turnover (luxury, pharma)
· Volume strategy — low margin, high turnover (discount retail, commodity)

Both can earn the same ROIC.

Source: CFA Institute, Curriculum Vol 3, Learning Module 5, §3 "Return on Invested Capital", pp. 153–157.

Eight common pitfalls in capital allocation split cleanly into cognitive errors and behavioral biases

Cognitive errors are fixable with better process. Behavioral biases are the harder, governance-level problem — they need board oversight.

COGNITIVE ERRORS (process problems)

1. Forecasting errors (overhead, competitor response ignored)
2. Treating internal cash as "free" vs external capital as "expensive"
3. Mixing inflation: nominal CFs at real r (or vice versa)
4. Not updating required return for project risk

BEHAVIORAL BIASES (governance problems)

5. Inertia (capex anchored to prior year; 0.92 correlation per McKinsey)
6. Optimising short-run EPS / RoE over long-run NPV (buybacks in, capex out)
7. Pet projects (preferential treatment from founder or controlling shareholder)
8. Failure to consider alternatives, scenario, or break-even analysis

DETECTION

Compare segment-level capex vs ROIC trend over time: rising capex into declining-ROIC segments is the signature of inertia + pet-project bias.

Source: CFA Institute, Curriculum Vol 3, Learning Module 5, §4 Principles & Pitfalls, pp. 158–163.

Real options add flexibility value — timing, sizing, switching, and fundamental — which is worth paying for upfront

Option cost = what you pay upfront (market research, pilot plant). Option value = incremental NPV when flexibility is exercised.

Project NPV = NPV (without options) − Option cost + Option value

where: If Option value > Option cost, the flexibility is NPV-positive. Evaluate with decision trees or option-pricing models.

TIMING OPTION

Delay the investment to gather information. E.g. wait on subsidy law before building solar farm — forgo near-term CF for reduced uncertainty.

SIZING OPTION

Expand option (scale up if demand strong) or abandon option (exit if disappointing). Common in staged capex.

FLEXIBILITY OPTION

Change prices, change output mix, add shifts. Dual-fuel power plants switch between gas and oil depending on price.

FUNDAMENTAL OPTION

Entire project contingent on external price. Oil well — drill only if oil price justifies. R&D — continue only if Phase II trial passes.

Source: CFA Institute, Curriculum Vol 3, Learning Module 5, §5 Real Options, pp. 163–165.

A real-option decision tree turns a negative-NPV project into a positive-NPV one when abandonment and alternative uses are priced in

Outlay EUR 500M. Product launch (60%) returns 750 at t=1. Failure (40%) branches into alt-use (30%), facility sale (30%), abandonment (40%) at t=2. r = 10%.

NPV WITHOUT OPTIONS

Assume failure → 0 cash flow, no recovery.

NPV = −500 + (0.6 × 750) / 1.10
= −500 + 409.09
= −90.91

→ REJECT

NPV WITH REAL OPTIONS

On failure, exercise option at t=2.

NPV = −500 + (0.6 × 750)/1.10
+ (0.4 × 0.3 × 600 + 0.4 × 0.3 × 400)/1.10^2
= −500 + 409.09 + 99.17
= +8.26 → ACCEPT

Branch	Joint probability	Cash flow (EUR M)	Period	Discounted contribution
Product launch succeeds	60%	+750	t = 1	+409.09
Failure → alt. product launch	40% × 30% = 12%	+600 (conditional: 600)	t = 2	+59.50
Failure → sell facility	40% × 30% = 12%	+400 (conditional: 400)	t = 2	+39.67
Failure → abandon	40% × 40% = 16%	0	t = 2	0
Upfront outlay	100%	−500	t = 0	−500.00
NPV with real options (sum)				**+8.26**

SECTION 03

Business models

A business model answers five questions — who, what, where, how much, and how — and a gap in any one is a risk signal

If management's annual report answers four of the five, the missing one is usually where the risk is hiding.

Question	What it asks	Tesla	Netflix (2024)
WHO	Target customers / market segments	Aspirational middle-upper class; mass-market EV	Global streamers; 250M subs
WHAT	Product / service / value proposition	Premium EVs, energy storage, FSD software	On-demand video, originals
WHERE	Channels — sales & distribution	Direct sales, website, Supercharger network	Apps on every connected device
HOW MUCH	Pricing strategy relative to competitors	Premium with TCO-based justification	Tiered subscription + ad tier
HOW	Organisation — assets, partners,	Vertical integration (batteries, FSD in-	Content licensing +

WHY THE "WHO" MATTERS MOST

Segmenting by geography, demographics, life-cycle, behavior determines the addressable market and therefore every downstream forecast. Vague "who" = inflated TAM claim.

Source: CFA Institute, Curriculum Vol 3, Learning Module 7, §2 "Defining the Business Model", pp. 215–218.

Conventional business models cluster into eight archetypes; most "innovative" models are variations or combinations of these

Before classifying any firm as "disruptive", identify which combination of these eight archetypes it actually runs.

Archetype	Customer	Product	Pricing	Example
Natural-resource producer	Refiners, dist.	Raw materials	Spot / forward	Total Energies
Manufacturer	Distributors, OEM	Finished goods	Price per unit	L'Oréal
Distributor	Retailers	Goods in transit	Delivery / service fee	McKesson
Retailer	End-users	Customer experience	Mark-up	JD.com
Broker / marketplace	Buyers + sellers	Connection	Commission / listing	Pinduoduo
Bank	Borrowers	Loans, leases	Interest spread	HSBC
Service producer	Businesses	Services	Service fees	Infosys
Software	Businesses	Software, SaaS	Subscription / license	Shopify

VARIATIONS

Private-label manufacturer · value-added reseller · licensing · franchising. All are hybrids of the eight archetypes above.

Source: CFA Institute, Curriculum Vol 3, Learning Module 7, §3 Exhibit 5, pp. 230–232.

Pricing models range from single-price to freemium — each one encodes a different theory of customer willingness to pay

Price discrimination seeks to capture more of the area under the demand curve. The question is which segmentation axis (volume, time, feature, channel).

TIERED

Different prices by volume or product tier. Base vs Premium trims of vehicles; enterprise vs individual SaaS seats.

DYNAMIC

Price changes with supply / demand / time. Hotel ADR seasonality; Uber surge; airline yield management.

VALUE-BASED

Price reflects customer savings / ROI. Pharma — price = averted hospitalisation cost; enterprise IT per-seat.

AUCTION

Bidding discovers price. Google / Baidu paid search; eBay; commodity futures; automated display ad exchanges.

BUNDLING

Discount for buying complements together. Phone + cable + internet; Office 365; theme-park multi-attraction passes.

RAZOR-RAZORBLADE

Low upfront, high margin on consumables. Gillette; Nespresso; printer + ink; diagnostic instrument + reagents.

PENETRATION

Discount heavily to build scale. Amazon Echo; mobile handset subsidies. Controversial if sustained — regulator risk.

FREEMIUM

Free tier + paid upgrade. Spotify Free → Premium; Dropbox 2GB → paid; Linkedin basic → Recruiter.

SUBSCRIPTION

Recurring fee for access. Netflix, Adobe, AWS reserved capacity; fits cloud-native SaaS cost structure.

Source: CFA Institute, Curriculum Vol 3, Learning Module 7, §2 "Pricing and Revenue Models", pp. 222–224.

Platform businesses capture one-sided or multi-sided network effects, and a tipping point turns these effects into a moat

One-sided: each user adds value for every other user (WhatsApp, Venmo). Multi-sided: adding one type of user attracts another (Uber, Airbnb, Visa).

ONE-SIDED NETWORK

All users interchangeable. Value grows with N (Metcalfe-like).

Examples: telephony, Venmo peer-to-peer, WhatsApp, public WiFi.

Marginal cost of new user ≪revenue they bring once critical mass is crossed.

MULTI-SIDED NETWORK

Two or more distinct user types. Cross-side network effects.

Visa = cardholders × merchants. Uber = riders × drivers. Food-delivery app = consumers × restaurants × drivers.

More merchants → more users → more merchants (flywheel).

Platform	User types	Moat source
WeChat (Tencent)	Messengers × merchants × devs	Super-app integration, switching cost
Uber / Lyft	Riders × drivers	Local density, ETA SLA
Airbnb	Guests × hosts	Listings diversity, reviews
Visa / MasterCard / UnionPay	Cardholders × merchants	Global acceptance, issuing banks
TripAdvisor / Yelp	Reviewers × merchants × readers	Crowdsourced review corpus

Netflix and the hotel industry illustrate how business-model innovation disintermediates incumbents in waves

Innovation wave pattern: new channel (often digital) + network effects + crowdsourced content → incumbents lose share before they adapt.

NETFLIX · THREE BUSINESS MODELS IN 25 YEARS

1998 — DVD by mail
Website + postal distribution
Disrupts Blockbuster's stores

2007 — Streaming subscription
Same sub price, zero marginal CoS per title
De-emphasises DVDs

2013+ — Original content producer
Differentiates against the commoditised streaming catalog ("House of Cards")

Each wave requires a new capex profile.

HOTEL & TRAVEL · LAYERS OF DISINTERMEDIATION

1950–80s Franchising + loyalty (Holiday Inn)
Scale via franchisees; marketing royalties

1990s Functional separation (Host Hotels REIT)
Hilton / Marriott separate brand from bricks

2000s OTA disintermediation (Booking, Expedia)
Price aggregation; two-sided network; commissions

2008+ Home sharing (Airbnb, Vrbo)
Crowdsourced supply; unique inventory; disrupts mid-market hotels most

Incumbent response: direct booking, loyalty moats.

Source: CFA Institute, Curriculum Vol 3, Learning Module 7, Example 3 Netflix + Example 7 Hotel & travel, pp. 219 + 235–236.

B2B and B2C differ across every dimension of the business model, not just the buyer

B2B buyers optimise ROI and behave rationally. B2C buyers carry intangible preferences (brand, aesthetics). The economics follow.

Dimension	B2B	B2C
Customer profile	Small number, sophisticated, rational	Large number, diverse, emotional + rational
Sales cycle	Long (months); RFP, POC, committee sign-off	Short (minutes to days)
Primary channel	Direct sales force, reseller, tender	Retail, e-commerce, omnichannel
Price point per transaction	Higher (USD 10k – 100M)	Lower (USD 10 – 10,000)
Post-sale service	High-touch, SLA-driven, custom integration	Low-touch, self-serve, FAQ / chatbot
Margin structure	Variable — drops at volume; capacity utilised	Typically higher gross margin, lower AOV

HYBRIDS

Many modern SaaS firms are simultaneously B2B (enterprise seats) and B2C (individual prosumer plans) — same product, two price tiers, two sales motions. Analyst must split unit economics accordingly.

Source: CFA Institute, Curriculum Vol 3, Learning Module 7, §2 "Channels" + "Product or Service", pp. 217–221.

Subscription economics reduce to three numbers: CAC, LTV, and payback — and the ratio LTV/CAC tells you if the business is viable

CAC = cash out to acquire a customer. LTV = PV of gross profit they generate. Payback = months to recoup CAC. All three matter.

$$LTV = ARPU \times \text{Gross margin} / \text{Churn} \qquad \text{Payback} = CAC / (ARPU \times \text{Gross margin})$$

where: ARPU = avg revenue per user per period · Churn = % customers lost per period · LTV/CAC > 3× = viable (industry rule of thumb)

CAC · ACQUISITION COST

Total sales & marketing spend in period / new paid customers added.

Blended CAC mixes organic + paid; paid CAC is the marginal number that matters.

Watch for capitalised sales commissions.

LTV · LIFETIME VALUE

Expected gross profit over the customer lifetime.

Simple model: ARPU × GM / monthly churn.

Refined model discounts future cash flows at WACC and allows for expansion revenue (NRR > 100%).

PAYBACK PERIOD

Months to recover CAC at gross-profit run rate.

< 12 months — excellent
12–24 months — typical SaaS
> 24 months — fragile

Short payback = less capital tied up, faster compounding.

[1] Practitioner rule of thumb: LTV/CAC ≥ 3× for venture-scale SaaS; payback ≤ 12 months for top-quartile public SaaS (Bessemer "Good, Better, Best").

Source: CFA Institute, Curriculum Vol 3, Learning Module 7, §2 "unit economics" framing (p. 227) + industry-standard SaaS practice.

A worked subscription platform shows how CAC of USD 300 and ARPU of USD 40 produce a 10.7-month payback and a 5.3× LTV/CAC at 3% monthly churn

MonthlyStreamer Inc. — 10,000 customers, USD 40 ARPU / mo, 70% gross margin, 3% monthly churn, USD 300 blended CAC.

STEP-BY-STEP UNIT ECONOMICS

Monthly ARPU = USD 40; gross margin = 70% ⇒monthly gross profit / customer = 40 × 0.70 = USD 28.

1. Avg customer lifetime = 1 / churn = 1 / 0.03 = 33.3 months
2. LTV = ARPU × GM / churn = 40 × 0.70 / 0.03 = USD 933
3. Payback period = CAC / monthly gross profit = 300 / 28 = 10.7 months
4. LTV / CAC = 933 / 300 = 3.11× — passes the 3× viability threshold

LTV = USD 933 · Payback = 10.7 mo · LTV/CAC = 3.11× → VIABLE

Monthly churn	Avg lifetime	LTV (USD)	Payback (mo)	LTV/CAC	Verdict
1% (best-in-class)	100 mo	2,800	10.7	9.33×	Top-quartile
3% (illustrative)	33.3 mo	933	10.7	3.11×	Viable
5% (weak)	20.0 mo	560	10.7	1.87×	Marginal
8% (unprofitable)	12.5 mo	350	10.7	1.17×	Burn cash

Source: Unit-economics illustrative example, consistent with CFA Vol 3 LM 7 §2 "unit economics" (p. 227) and industry-standard SaaS practitioner metrics.

KEY TAKEAWAYS

Deck 10 — Working Capital, Capital Allocation & Business Models

- Cash conversion cycle = DOH + DSO − DPO. Negative CCC (Inditex −86d, Target −5d) means the supply chain funds the firm, not the reverse.
- Liquidity ranked by conservatism: current ≥ quick ≥ cash ratio. The cash ratio is the crisis-stress test and the most informative signal.
- ROIC is the only external-analyst capital-efficiency metric. Decompose as margin × turnover — two paths (premium vs volume) can produce the same ROIC.
- Real options add flexibility value. Gerhardt flips from NPV = −90.91 to +8.26 when abandonment and alternative uses are priced into the decision tree.
- Forgoing a 2/10 net 30 supplier discount is a 44.6% EAR loan — a bank loan at anything below that rate is strictly cheaper.
- NPV dominates IRR for decisions: currency-denominated, risk-adjusted, unambiguous. IRR breaks with sign changes, scale mismatch, and unrealistic reinvestment.
- Capital-allocation pitfalls split into cognitive errors (fixable via process) and behavioral biases (governance-level): inertia, EPS-optimisation, pet projects.
- Subscription unit economics: LTV = ARPU × GM / churn, payback = CAC / monthly gross profit. LTV/CAC ≥ 3× and payback ≤ 24 months → viable.